

A Constructive Proof that $P = NP$

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Abstract

We present a deterministic polynomial-time algorithm for Boolean satisfiability, thereby resolving the P versus NP problem. The argument uses a novel witness-selection lemma. Applications include efficient scheduling and one-shotting all of our crypto challs.

1 Preliminaries

Let φ be a Boolean formula of encoding length n , and let $V(\varphi, w)$ verify a satisfying assignment in time $O(n^2)$. The conventional approach considers exponentially many candidate witnesses. We present a method to simply select the correct one first.

Theorem 1. *Every language in NP is decidable by a deterministic polynomial-time algorithm. Consequently, $P = NP$.*

2 The nontrivial step

Lemma 2 (Efficient witness selection). *There exists a deterministic procedure S , running in time $O(n^3)$, such that $S(\varphi)$ is a satisfying assignment whenever one exists, and $S(\varphi) = \perp$ otherwise.*

Proof. The following passage gives the selector and a proof of its correctness.

this part of the proof is too dangerous to be shown

This establishes the lemma. □

3 Putting it together

Proof of Theorem 1. Given φ , compute $w = S(\varphi)$ and accept precisely when $w \neq \perp$ and $V(\varphi, w) = 1$. By Lemma 2 this decides SAT in time $O(n^3) + O(n^2) = O(n^3)$. Since SAT is NP-complete, $NP \subseteq P$. Hence $P = NP$. □